

Robust Newton–Raphson Power Flow Analysis for Singular and Ill-Conditioned Systems

Final Presentation

2105094 2105102 2105104 2105116 2105117

Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology

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Outline

- 1 Motivation
- 2 Methods
- 3 Implementation and Validation
- 4 Results
- 5 Discussion and Conclusion

Power Flow with Newton–Raphson

- Power flow finds every bus voltage from the given loads and generation. The equations are nonlinear, $F(x) = 0$.
- State x : real and imaginary voltage parts (e, f) at every non-slack bus-phase.

Newton–Raphson iteration

$$J_k \Delta x_k = -F(x_k), \quad x_{k+1} = x_k + \Delta x_k, \quad J_k = \left. \frac{\partial F}{\partial x} \right|_{x_k}$$

- Each iteration solves one linear system with the Jacobian J .
- **That linear solve is the only numerically delicate step**, and it is the only thing we change in this project.

Why the Jacobian Becomes Singular

Physical cause

- Delta and ungrounded windings block zero-sequence current.
- A region with no path to ground has an *undetermined* common-mode voltage V_0 .
- This gives a null direction in J : exactly singular, or $\kappa_2 \sim 10^{17}$ – 10^{18} .

Numerical consequence

- LU reports an error only on an *exactly* zero pivot.
- Otherwise the error grows to about $\kappa_2(J) \varepsilon_{\text{mach}}$, with $\varepsilon_{\text{mach}} \approx 10^{-16}$.
- The residual $\|J\Delta x + F\|$ can still look small, so **the answer is silently wrong**.

Base Paper and Our Questions

Base paper: Jang, Kim & Kim, *IEEE Access*, 2023

- Replaces J^{-1} with the SVD pseudo-inverse J^{+} when J is singular.
- Tested on IEEE 4-, 13-, 37- and 69-bus feeders.
- The 13-bus error ($\approx 2 \times 10^{-4}$ pu) is much larger than the others and is *not explained*.

Our questions

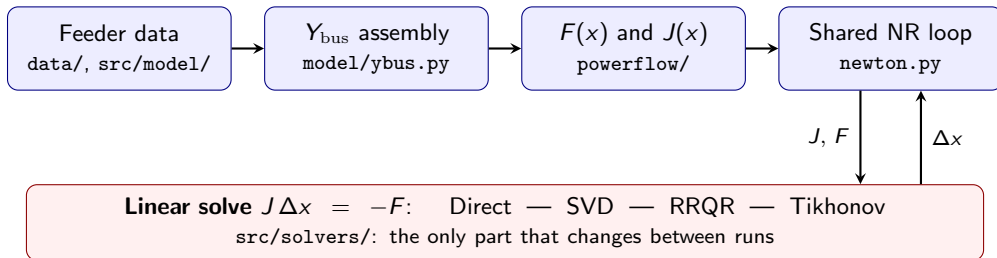
- 1 Is the pseudo-inverse the best remedy, or does the answer depend on the feeder?
- 2 How do rank-revealing QR and Tikhonov regularization compare, inside the same NR solver?
- 3 Why is the 13-bus error larger?

Four Linear Solvers, One NR Loop

Method	Step computed	Key detail
Direct	LU solve of $J\Delta x = \text{rhs}$	Rejects the step if $\kappa_2(J) > 10^{12}$.
SVD (base paper)	$\Delta x = V\Sigma^+U^H \text{rhs}$	Sets $1/\sigma_i \rightarrow 0$ for $\sigma_i \leq \tau = \text{rtol} \cdot \sigma_{\max}$.
RRQR	Pivoted QR $JP = QR$, then LAPACK GELSY	Minimum-norm least squares. Plain QR with back-substitution breaks when R is singular.
Tikhonov	$\min \ J\Delta x - \text{rhs}\ ^2 + \lambda \ \Delta x\ ^2$	Augmented system $\begin{bmatrix} J \\ \sqrt{\lambda}I \end{bmatrix}$, $\lambda = \alpha \sigma_{\max}^2$; reported feeder runs explicitly use $\alpha = 10^{-8}$. Never forms $J^T J$, which would square κ .

- SVD and RRQR take the **exact minimum-norm** step.
- Tikhonov takes a **damped** step: a small bias in exchange for stability.

Shared Architecture



- Every run uses the same NR loop, data and tolerance. Only the `method` argument changes.
- Python with NumPy, SciPy and pandapower. OpenDSS provides the reference solutions.
- 86 automated tests (pytest), and every experiment script runs on its own.

Test Feeders

Feeder	Modification	n	$\kappa_2(J_0)$	Near-0 σ_i
4-bus	Yg- Δ , no secondary load	18	$61.5 \rightarrow > 10^{12}$	0
13-bus	T1 $\rightarrow$ Yg- Δ , T2 $\rightarrow$ Δ - Δ	64	$\sim 10^{18}$	2
37-bus (radial)	Official Δ - Δ , unloaded XFM1	222	$\sim 10^{18}$	4
37-bus (quasi-radial)	+ three loop-closing lines	222	$\sim 10^{18}$	4
69-bus	$3 \times$ Yg- Δ (3-4, 3-28, 3-36)	408	2.3×10^5	0

- n is the number of real unknowns, and $\kappa_2 \sim 10^{18}$ means machine-singular.
- All feeders were rebuilt from official IEEE, OpenDSS and MATPOWER data, with every assumption recorded in `data/provenance.yml`.
- The IEEE 118-bus system (236–1888 unknowns) is used only as a runtime benchmark.

1. Finite-difference Jacobian check

$$E_J = \frac{\|J^{\text{analytic}} - J^{\text{FD}}\|_F}{\|J^{\text{FD}}\|_F} < 10^{-9}$$

on every feeder, at a perturbed point (not the flat start).

2. OpenDSS cross-check

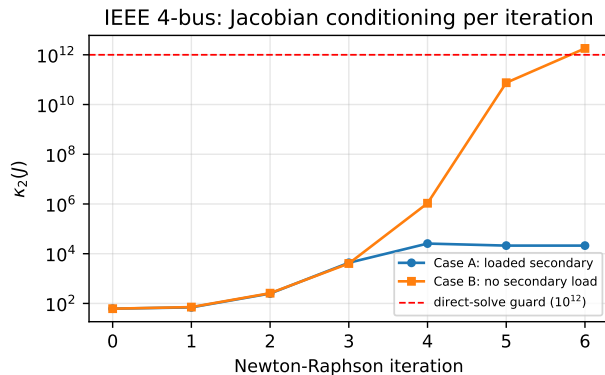
The same circuit is solved independently in OpenDSS and the voltages are compared.

Two real bugs caught

- 13-bus: loads $3\times$ too light, because they were divided by the three-phase base.
- 69-bus: loads $3\times$ too heavy, because each phase got the total load.

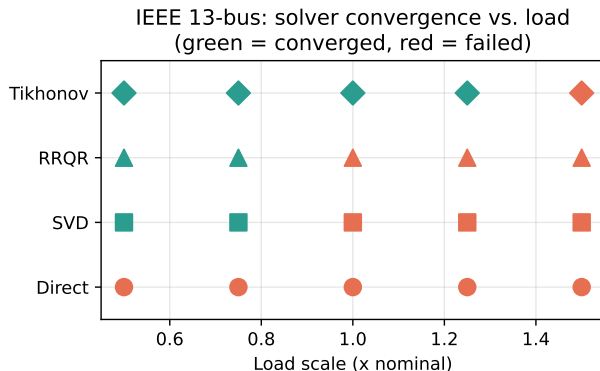
Both were found by first checking a *small, well-conditioned sub-circuit*. On the full system the 69-bus bug looked like a plausible 33% error at one bus.

IEEE 4-Bus: Reproducing the Failure



- Yg- Δ transformer (6 MVA, 12.47/4.16 kV) with no secondary load.
- κ_2 rises from 61.5 to 1.8×10^{12} by iteration 7, so the direct guard stops.
- SVD, RRQR and Tikhonov **all converge**.
- Loaded case: $\kappa_2 \leq 2.6 \times 10^4$, and the solution matches OpenDSS to 1.5×10^{-6} pu.

IEEE 13-Bus: Minimum-Norm Steps Oscillate



- The whole downstream network floats, giving 2 near-zero singular values (T1 and bus 634).
- Direct fails by construction.
- SVD and RRQR converge at $0.5\times$ and $0.75\times$ load but **oscillate from $1.0\times$** .
- Only Tikhonov converges at nominal load (15 iterations), up to $1.25\times$.

Here, damping is needed for convergence at all, not only to handle the singularity.

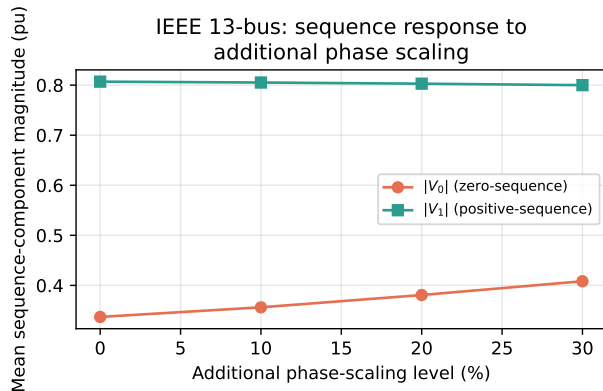
IEEE 13-Bus: α Must Be Tuned

α	Converged	Iterations	Final $\ F\ _\infty$
0	×	80	1.93×10^{-2}
10^{-12}	×	80	1.64×10^{-2}
10^{-10}	×	80	1.50×10^{-2}
10^{-8}	✓	15	2.02×10^{-10}
10^{-6}	×	80	1.80×10^{-3}
10^{-4}	×	80	1.22×10^{-2}
10^{-2}	×	80	3.22×10^{-1}

Sensitivity result

10^{-8} is the only convergent point on this grid. Too little damping leaves the oscillation; too much damping stalls the Newton iteration.

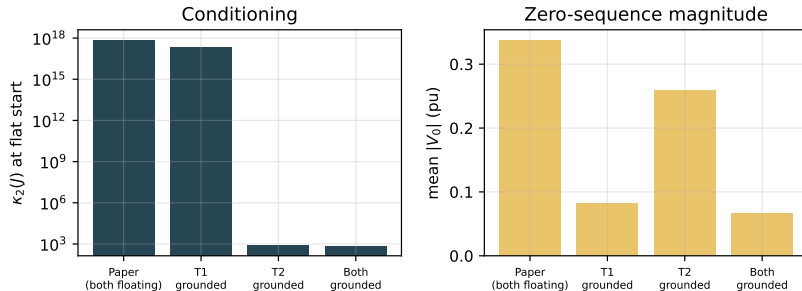
IEEE 13-Bus: Imbalance Goes into V_0



- Extra imbalance: phase $a \times (1 + L)$ and phase $c \times (1 - L)$, for $L = 0\text{--}30\%$.
- $|V_0|$: $0.337 \rightarrow 0.408$ pu (+21%).
- $|V_1|$: $0.807 \rightarrow 0.800$ pu (under 1%).
- The imbalance goes into the component the equations *cannot determine*.

IEEE 13-Bus: Grounding the Transformers

IEEE 13-bus: effect of grounding the floating transformers



- Both transformers grounded: κ_2 falls from $\sim 10^{18}$ to ≈ 773 , and $|V_0|$ from 0.337 to 0.066 pu.
- Grounding T2 alone removes the extreme conditioning. Grounding T1 alone gives most of the drop in V_0 .

37-bus: the reversal

- 4 near-zero singular values, $\kappa_2 \sim 10^{18}$.
- SVD and RRQR converge in **10 iterations** with no oscillation.
- Tikhonov with the same $\alpha = 10^{-8}$ **fails** within 80 iterations.
- The quasi-radial variant shows the same pattern.

69-bus: ill-conditioned but not singular

- κ_2 : $2.3 \times 10^5 \rightarrow 5.7 \times 10^{10}$, which stays below the guard.
- Direct, SVD and RRQR converge in 7 iterations. Tikhonov fails.
- Y_{bus} has an exact null direction, but J does not, because the load in the floating region removes the degeneracy.

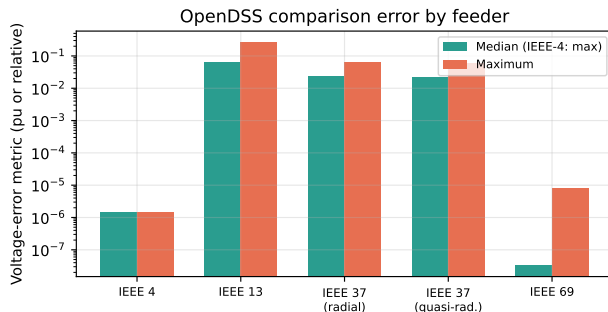
Summary: No Single Winner

Feeder (nominal load)	Direct	SVD	RRQR	Tikhonov
4-bus, loaded secondary	✓	✓	✓	✓
4-bus, no secondary load	✗	✓	✓	✓
13-bus	✗	✗ oscillates	✗ oscillates	✓
37-bus (radial)	✗	✓	✓	✗
37-bus (quasi-radial)	✗	✓	✓	✗
69-bus	✓	✓	✓	✗

Main finding

No single solver, and no single fixed regularization strength, is best on every feeder. The values of κ_2 and the null-space size do *not* predict which method converges.

Cross-Validation with OpenDSS

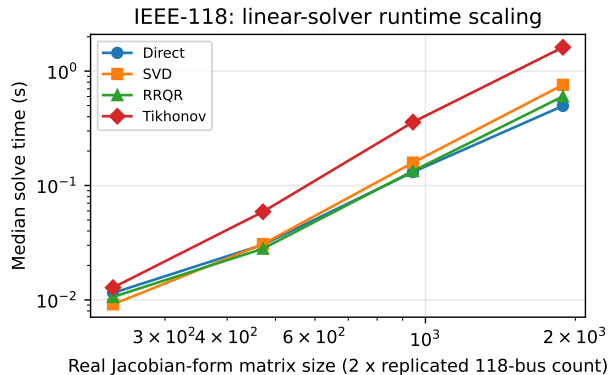


Feeder	Median	Max
4-bus*	—	1.5×10^{-6}
13-bus	6.6%	26.5%
37 radial	2.3%	6.5%
37 quasi-rad.	2.3%	6.1%
69-bus	3.3×10^{-8}	8.2×10^{-6}

* Absolute error in pu. The other rows give relative line-to-line error.

- 4- and 69-bus match almost exactly, which confirms the model and the per-unit conversion.
- On 13- and 37-bus, two things add up: the ambiguity in the null space and our wye approximation of delta loads.

IEEE 118-Bus: Runtime Scaling



- Real case118 Y_{bus} , repeated up to $8\times$, gives 236–1888 unknowns.
- Fitted runtime exponents are about 1.8–2.4, below the cubic asymptotic cost because of BLAS and fixed overhead.
- Tikhonov is always slowest, because its augmented system has twice the rows.
- Direct is fastest at the larger sizes.

- ① **Choose the solver to fit the rank deficiency.** The pseudo-inverse is not a universal fix.
- ② **Exact minimum-norm steps can oscillate.** Damping (Tikhonov) helps, but α has to be tuned for each feeder.
- ③ **A floating network has no unique solution.** Solvers and OpenDSS disagree mainly in V_0 , which partly explains the base paper's larger 13-bus error.
- ④ **Check small, well-conditioned pieces first.** On the full ill-conditioned systems, real bugs look like normal disagreement.

Limitations and Future Work

Limitations

- Delta loads are modelled as wye equivalents, regulators are ideal, and ungrounded wye is not implemented.
- No line search or trust region, and α is fixed.
- We defined our own imbalance rule, and the 69-bus feeder is balanced only.
- The base paper is ambiguous about the 13-bus load count, the 37-bus reactance and the 69-bus voltage base.

Future work

- Add line search or trust-region globalization.
- Choose α automatically (L-curve, discrepancy principle).
- Model delta loads and ungrounded-wye windings exactly.
- Use sparse factorizations for large systems.

- We built one shared three-phase NR solver and compared Direct, SVD, RRQR and Tikhonov on five feeder variants plus a 118-bus benchmark.
- **No method is best everywhere:**
 - 13-bus: only Tikhonov converges.
 - 37-bus: SVD and RRQR converge and Tikhonov fails.
- On a floating network, load imbalance goes into the undetermined zero-sequence voltage, which partly explains the base paper's 13-bus error.
- Checking small pieces against OpenDSS found two real per-unit bugs.

References

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Thank You

Questions?